

Cell-mediated and serology-based tests for *Mycobacterium ulcerans* disease: A systematic review and meta-analysis - Study Protocol

Citation

Cell-mediated and serology-based tests for *Mycobacterium ulcerans* disease: A systematic review and meta-analysis.

*Michael S. Avumegah^{1,2,3}; Nilakshi T. Waidyatillake^{4,5}; Wojtek P. Michalski⁶; Daniel P. O'Brien^{4,7}; Tiffanie M. Nelson^{2,3}; Eugene Athan^{2,3,4}

1. The University of Queensland, School of Chemistry and Molecular Bioscience, Brisbane, Australia.
2. Deakin University, School of Medicine, Geelong Australia.
3. Geelong Centre for Emerging Infectious Diseases (GCEID), Geelong, Australia.
4. Department of Infectious Diseases, Barwon Health, Geelong, Australia.
5. Allergy and Lung Health Unit, Melbourne School of Population and global Health, The University of Melbourne, Melbourne, Australia.
6. Commonwealth Scientific and Industrial Research Organisation, Australian Animal Health Laboratory (CSIRO AAHL), Geelong, Australia
7. Department of Medicine and Infectious Diseases, Royal Melbourne Hospital, The University of Melbourne, Melbourne, Australia.

* **Corresponding author:** s.avumegah@uq.edu.au

The University of Queensland, School of Chemistry and Molecular Biosciences, Brisbane, Australia.

Review question

What attempts have been made to develop cell-mediated and/or serology-based tests for *Mycobacterium ulcerans* disease within at risk population?

Context and rationale

Current methods of diagnosis of the Buruli ulcer disease include, polymerase chain reaction, histopathology and culture of the pathogen. These methods have their infrastructural and resource challenges. Cell-mediated and/or serology-based tests have been suggested as easy and quick to potentially aid diagnosis of the disease.

Searches

The following online reference databases were searched for relevant articles: PubMed (1930 to June 2019), Web of Science / ISI Web of Knowledge (1930 to June 2019) and the Buruli ulcer disease database maintained by WHO in Geneva, Switzerland (<http://www.who.int/buruli/en>).

Search strategy

This study collated all research attempts made in developing cell-mediated immunity and/or serology-based tests for *M. ulcerans* disease by reviewing all published studies until June 2019 that have reported relevant efforts. The search terms used were: Buruli ulcer and/or *Mycobacterium ulcerans* in combination with the following terms: “cell-mediated screening test”, “serological screening test” and “diagnostic tests”. All searches were with no language restrictions.

Inclusion criteria:

Study papers that met selection criteria, included

1. Original full text papers and conference abstracts.
2. Studies that had attempted developing protein/antigen preparations either cell-derived or recombinant from *M. ulcerans* and had utilized them in cell-mediated and/or serology-based test for *M. ulcerans* disease.

Exclusion criteria:

Reviews papers were excluded.

Human disease modelled

Buruli ulcer/*M. ulcerans* disease.

Main outcomes(s)

Summary effect of the usefulness of cell-mediated and serology-based tests as tools for *M. ulcerans* disease.

Data extraction

The following study information were extracted and assessed;

1. Country and year of study.
2. *M. ulcerans* strains used for antigen preparation.
3. Culture media used for pathogen growth.
4. Method of protein preparation (Heat and/or non-heat treatment of pathogen).
5. Pathogen cell-derived (lysate/culture filtrate) and/or recombinant proteins used as test antigens or stimulant.
6. Immuno-detection method employed (ELISA, western blot, etc.).
7. Samples tested (Blood and/or serum) and were they human or animals?
8. Outcome (Sensitivity and specificity results).

Data collation followed the guidelines for review structure in the PRISMA checklist. The extracted data was cross-checked with a second reviewer and any discrepancies were resolved by a senior author.

Quality assessment of studies

The Newcastle-Ottawa Scale (NOS) was used to assess the quality of the included studies. NOS is used to assess the quality of non-randomised studies, and scores as follows: **Good quality score:** 3 or 4 stars in selection domain and 1 or 2 stars in comparability domain and 2 or 3 stars in outcome/exposure domain. **Fair quality score:** 2 stars in selection domain and 1 or 2 stars in comparability domain and 2 or 3 stars in outcome/exposure domain. **Poor quality score:** 0 or 1 star in selection domain or 0 star in comparability domain or 0 or 1 star in outcome/exposure domain.

Strategy for data synthesis

Summary effect was generated using the positivity and negativity outcomes of each selected article in relation to test antigen(s).

Statistical and meta-analyses

Studies that provided sufficient information, effect size (diagnostic odd ratio) with standard error (SE) were calculated and assigned weight at 95% confidence interval (CI). STATA statistical tool was then used to conduct a random-effects model meta-analysis to assess study heterogeneity and bias. The meta-analysis results were graphically displayed as forest and funnel plots ($p < 0.05$).

Comparator(s)/control.

Immune responses among healthy endemic controls and non-endemic controls using test antigens.

Participants/population

Buruli ulcer patients and healthy controls around the world as reported in relevant research articles.